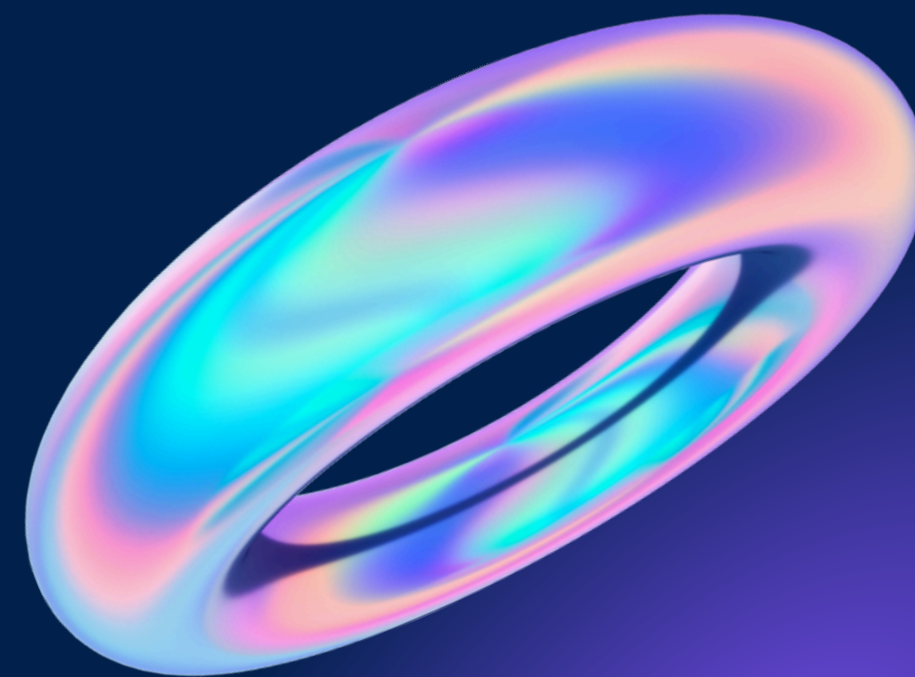


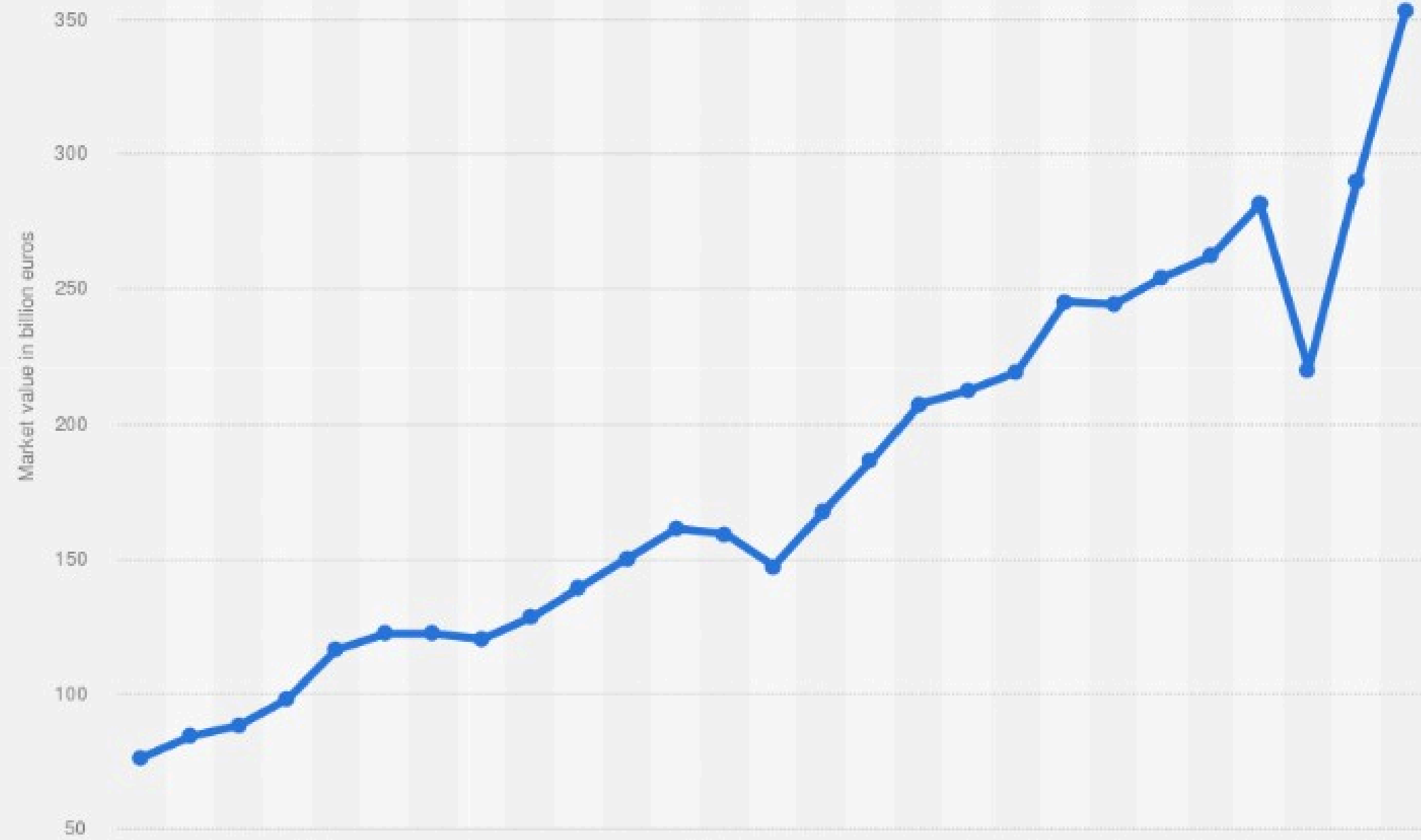


奢侈品業永續發展趨勢分析

中文四 陳靖誼
教育四 黃于庭
財政四 徐祥智

英文三 嚴維伶
經濟二 王玟晴
政治一 柯知言



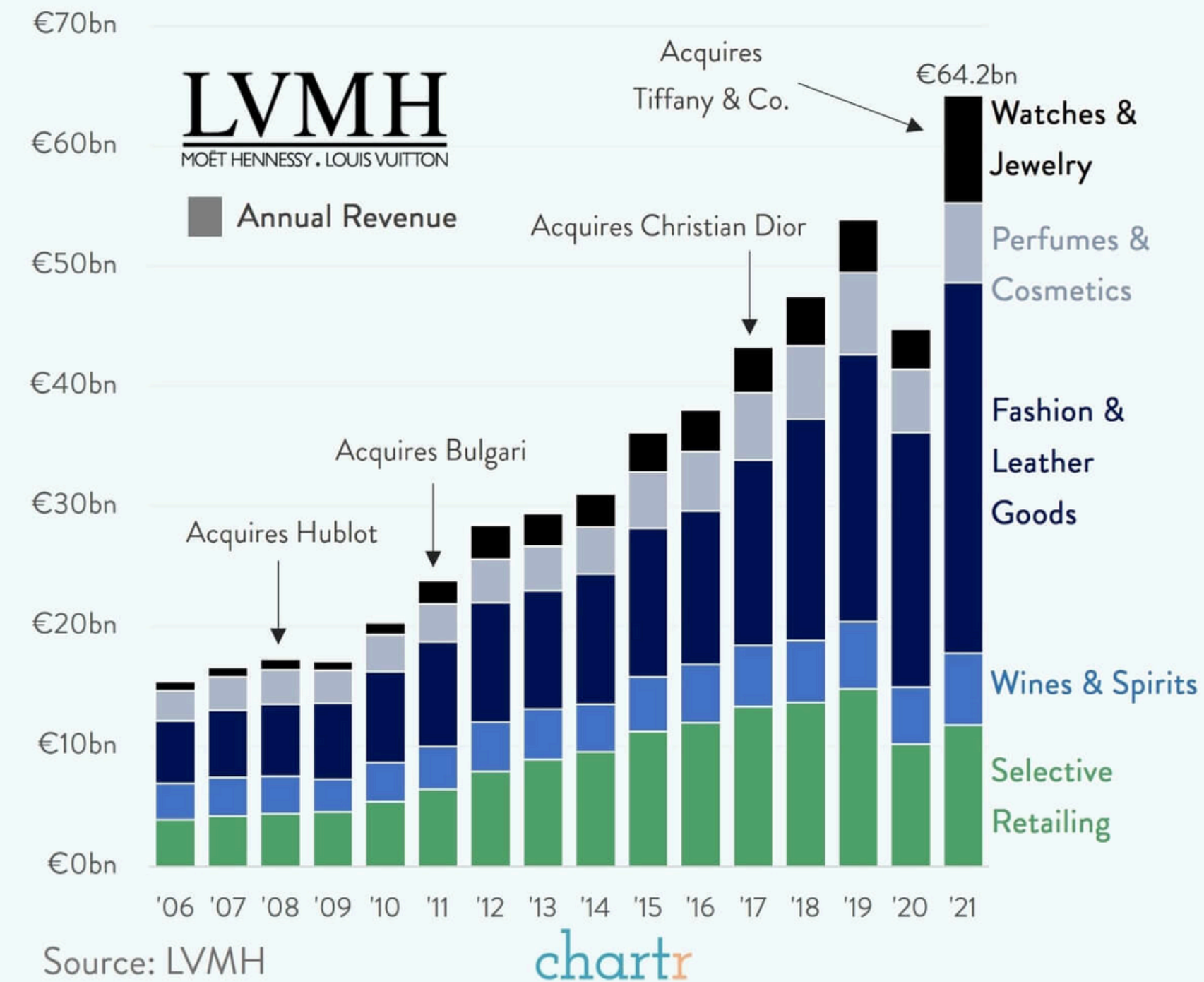


世界排名第三的製造業、總碳排佔10%



2012環境倡議計畫

Deal By Deal: How LVMH Became The World's Largest Luxury Co.



- 永續發展納入公司發展戰略

LIFE2020

1. 採購鏈應用最高標準
2. 改善場所環境性能指標
3. 減少二氧化碳排放
4. 所有產品的環境性能

LIFE360

1. 保護生物多樣性
2. 應對氣候變化
3. 循環經濟
4. 產品信息透明度



研究目標

- 奢侈品企業不僅在經濟方面具高度成長，也積極推動數位轉型與 ESG 永續發展的戰略計畫。
- 使用官方公佈的2016-2022年的碳排放數據預測LVMH在2023-2027年間的碳排放模型，探討該永續發展計畫的未來趨勢與成效。

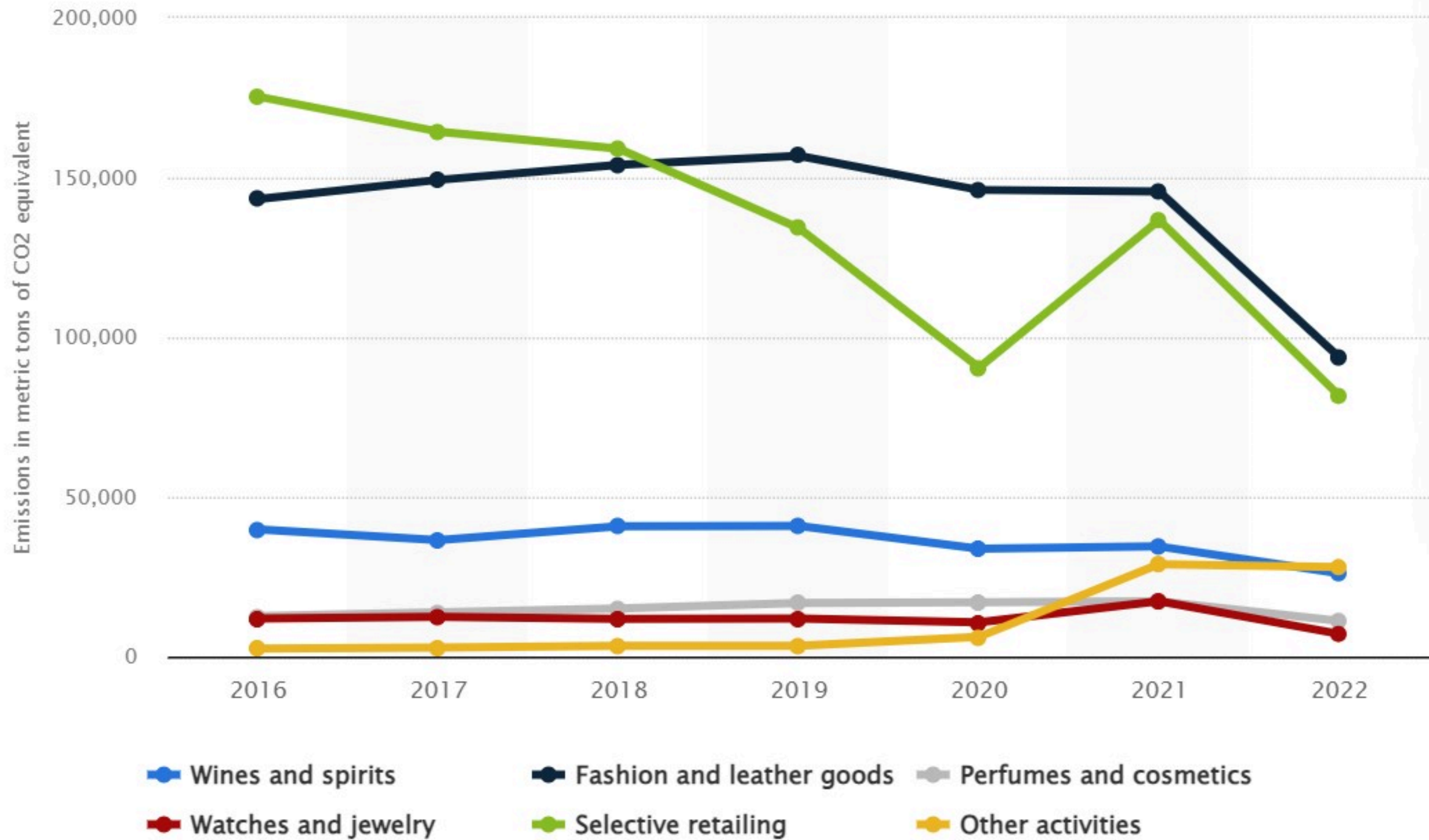




研究資料

- ✕ LV 2016-2022年的碳排放數據
 - 總碳排放量（噸）（Carbon Emissions (tons)）
- ✕ LV 2016-2022年各個分項數據
 - 時裝及皮革製品（Fashion & Leather Goods）
 - 精品零售（Selective Retailing）
 - 葡萄酒及烈酒（Wines & Spirits）
 - 鐘錶及珠寶（Watches & Jewelry）
 - 香水及化妝品（Perfumes & Cosmetics）
 - 其他活動與業務（Other Activities）





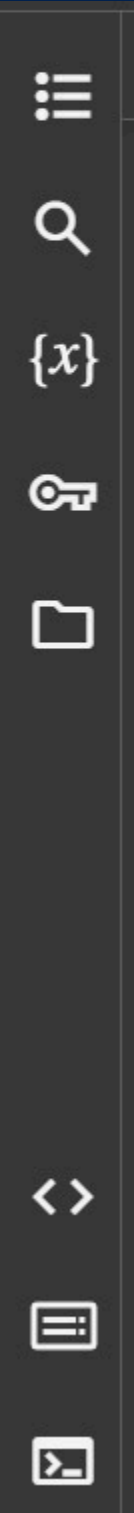
LV 2016-2022年碳排放數據



研究方法

將資料整理成結構化格式(.CSV)，包括總碳排放量、各個分項業務的碳排放量。之後將其分別匯入Python和Rapidminer中進行線性迴歸分析與時間序列分析，預測2023～2027每年碳排放變化的趨勢，並以此結果來評估LV未來的永續發展趨勢與其life360計畫能否成功。





+ 程式碼 + 文字

連線 Gemini ^



```
file_path <- '/content/未命名的試算表 - 工作表1.csv'

data <- read.csv(file_path)
```

```
[ ] install.packages("ggplot2")
install.packages("forecast")
library(ggplot2)
library(forecast)
```

Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

Installing package into ‘/usr/local/lib/R/site-library’
(as ‘lib’ is unspecified)

```
[ ] head(data)
```

A data.frame: 6 × 8

	Year	Carbon_Emissions	Fashion_Leather.Goods	Selective.Retailing	Wines_Spirits	Watches_Jewelry	Perfumes_Cosmetics	Other_Activities
	<int>	<dbl>	<int>	<int>	<int>	<int>	<int>	<int>
1	2016	385629	143336	175377	39787	11950	12650	2529




```
[ ] Call:
lm(formula = Carbon_Emissions ~ Year, data = data)

Residuals:
    1     2     3     4     5     6     7
-16387 -5150 17237 14628 -27440 66121 -49009

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 35806072   15576663   2.299   0.0699 .
Year        -17562     7715   -2.276   0.0719 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 40820 on 5 degrees of freedom
Multiple R-squared:  0.5089,    Adjusted R-squared:  0.4107
F-statistic: 5.181 on 1 and 5 DF,  p-value: 0.07187
```

```
[ ] # 創建一個包含2023到2027年的數據框
future_years <- data.frame(Year = 2023:2027)

# 使用模型進行預測
predictions <- predict(model, newdata = future_years)
future_data <- cbind(future_years, Carbon_Emissions = predictions)
print(future_data)
```



```
[ ] # 創建一個包含2023到2027年的數據框
future_years <- data.frame(Year = 2023:2027)

# 使用模型進行預測
predictions <- predict(model, newdata = future_years)
future_data <- cbind(future_years, Carbon_Emissions = predictions)
print(future_data)
```

```
↔ Year Carbon_Emissions
1 2023      279085.6
2 2024      261524.0
3 2025      243962.5
4 2026      226401.0
5 2027      208839.4
```

```
[ ] #預測走向折線圖(Carbon_Emissions)
ggplot() +
  geom_point(data = data, aes(x = Year, y = Carbon_Emissions), color = "blue") +
  geom_line(data = data, aes(x = Year, y = Carbon_Emissions), color = "blue") +
  geom_point(data = future_data, aes(x = Year, y = Carbon_Emissions), color = "red") +
  geom_line(data = future_data, aes(x = Year, y = Carbon_Emissions), color = "red") +
  labs(title = "數據預測",
        x = "年份",
        y = "值") +
```



研究結果

Year Fashion_Leather.Goods

1	2023	117839.43
2	2024	111980.25
3	2025	106121.07
4	2026	100261.89
5	2027	94402.71

Year Selective.Retailing

1	2023	76666.57
2	2024	62190.46
3	2025	47714.36
4	2026	33238.25
5	2027	18762.14

Year Wines_Spirits

1	2023	28618.57
2	2024	26760.79
3	2025	24903.00
4	2026	23045.21
5	2027	21187.43

Year Watches_Jewelry

1	2023	11016.29
2	2024	10801.75
3	2025	10587.21
4	2026	10372.68
5	2027	10158.14

Year Perfumes_Cosmetics

1	2023	15436.43
2	2024	15587.86
3	2025	15739.29
4	2026	15890.71
5	2027	16042.14

Year Other_Activities

1	2023	29508.29
2	2024	34202.93
3	2025	38897.57
4	2026	43592.21
5	2027	48286.86

Year Carbon_Emissions

1	2023	279085.6
2	2024	261524.0
3	2025	243962.5
4	2026	226401.0
5	2027	208839.4



Views:

Design

Results

Turbo Prep

Auto Model

Interactive
Analysis

Find data, operators...etc

All Studio

Repository

Import Data

- Titanic-Train (4/22/24 9:22 PM - 135
- train (4/22/24 5:49 PM - 353 kB)
- train3 (5/20/24 6:38 PM - 19.6 MB)
- 專案 (5/22/24 5:47 AM - 3 kB)
- 專案_2 (5/23/24 9:28 AM - 3 kB)
- 專案_3 (5/27/24 4:10 AM - 4 kB)

processes

20240506white spirit (5/19/24 2:27 A

Operators

FORECAST

Modeling (7)

Time Series (7)

Forecasting (6)

Apply Forecast

ARIMA

We found "Forecasting", "LifeStyle Marketing" and one more result in the Marketplace. [Show me!](#)

Process

Process

67%

Retrieve 專案_3



Retrieve 專案_3 (2)



Retrieve 專案_3 (3)



Retrieve 專案_3 (4)



Retrieve 專案_3 (5)



Retrieve 專案_3 (6)



Retrieve 專案_3 (7)



ARIMA



ARIMA (2)



ARIMA (3)



ARIMA (4)



ARIMA (5)



ARIMA (6)



Apply Forecast



Apply Forecast (2)



Apply Forecast (3)



Apply Forecast (4)



Apply Forecast (5)



Apply Forecast (6)



Leverage the Wisdom of Crowds to get operator recommendations based on your process design!

Activate Wisdom of Crowds

Parameters

Process

logverbosity

init

logfile

resultfile

random seed

2001

[Hide advanced parameters](#)

[Change compatibility \(10.3.001\)](#)

Help

Process

RapidMiner Studio Core

Synopsis

The root operator which is the outer most operator of every process.

Description

Each process must contain exactly one operator of this class, and it must be the root operator of the process.

ARIMA模型（英語：**A**utoregressive **I**ntegrated **M**oving **A**verage model），差分整合移動平均自我迴歸模型，又稱整合移動平均自我迴歸模型（移動也可稱作滑動），為**時間序列**預測分析方法之一。ARIMA（ p, d, q ）中，AR為**自我迴歸**， p 為自我迴歸項數；MA為移動平均， q 為滑動平均項數， d 為使之成為平穩序列所做的差分次數（階數）。「差分」一詞雖未出現在ARIMA的英文名稱中，卻是使時間序列得以平穩關鍵的步驟。

ARIMA（ p, d, q ）模型是**ARMA**（ p, q ）模型的擴展。ARIMA（ p, d, q ）模型可以表示為：

$$\left(1 - \sum_{i=1}^p \phi_i L^i\right) (1 - L)^d X_t = \left(1 + \sum_{i=1}^q \theta_i L^i\right) \varepsilon_t$$

其中 L 是滯後算子（Lag operator）， $d \in \mathbb{Z}, d > 0$

模型特點 [編輯]

- 不直接考慮其他相關隨機變數的變化。

ARIMA模型運用的流程 [編輯]

- 根據**時間序列**的散點圖、自我相關函數和偏自我相關函數圖識別其平穩性。
- 對非平穩的時間序列數據進行平穩化處理。直到處理後的**自我相關函數**和**偏自我相關函數**的數值非顯著非零。
- 根據所識別出來的特徵建立相應的時間序列模型。平穩化處理後，若**偏自我相關函數**是**截尾**的，而**自我相關函數**是**拖尾**的，則建立**AR模型**；若**偏自我相關函數**是**拖尾**的，而**自我相關函數**是**截尾**的，則建立**MA模型**；若**偏自我相關函數**和**自我相關函數**均是**拖尾**的，則序列適合**ARMA模型**。
- 參數估計，檢定是否具有統計意義。
- 假說檢定，判斷（診斷）**殘差序列**是否為**白噪聲序列**。
- 利用已通過檢定的模型進行預測。



Views:

Design

Results

Turbo Prep

Auto Model

Interactive
Analysis

Find data, operators...etc

All Studio

Repository

+ Import Data

- smoking (5/21/24 1:20 AM - 8 kB)
- test (5/5/24 10:19 AM - 143 kB)
- train (5/5/24 10:18 AM - 353 kB)
- 專案1 (5/22/24 5:50 PM - 7 kB)
- 專案2 (5/22/24 10:27 PM - 3 kB)
- 專案3 (5/27/24 4:28 AM - 4 kB)
- titanic (4/24/24 3:50 AM - 10 kB)

DB (Legacy)

Operators

arima

- Modeling (3)
 - Time Series (3)
 - Forecasting (2)
 - Apply Forecast
 - ARIMA
 - Utility (1)
 - Generate Data (ARIMA)

No results were found.

Process

Process

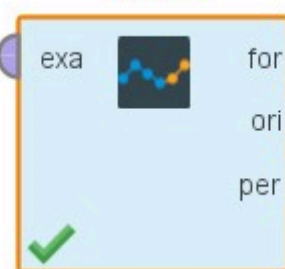
Process

inp

Retrieve 專案_3

ARIMA

Apply Forecast



Leverage the Wisdom of Crowds to get operator recommendations based on your process design!

Activate Wisdom of Crowds

Parameters

ARIMA

indices attribute

Year

☒ sort time series

p: order of the autoregressive model

1

d: degree of differencing

0

q: order of the moving-average model

0

[Hide advanced parameters](#)

Help

ARIMA
Time Series

Tags: [Time Series](#), [Forecast](#), [Arima](#), [Arma](#), [Ar](#), [Ma](#), [Forecasting](#)

Synopsis

This operator trains an ARIMA model for a selected time series attribute.

[Jump to Tutorial Process](#)

Description

Repository

Import Data

- smoking (5/21/24 1:20 AM - 8 kB)
- test (5/5/24 10:19 AM - 143 kB)
- train (5/5/24 10:18 AM - 353 kB)
- 專案1 (5/22/24 5:50 PM - 7 kB)
- 專案2 (5/22/24 10:27 PM - 3 kB)
- 專案3 (5/27/24 4:28 AM - 4 kB)
- titanic (4/24/24 3:50 AM - 10 kB)

DB (Legacy)

Operators

arma

- Modeling (3)
 - Time Series (3)
 - Forecasting (2)
 - Apply Forecast
 - ARIMA
 - Utility (1)
 - Generate Data (ARIMA)

No results were found.

Process

Process

Process

Retrieve 專案_3

ARIMA

Apply Forecast

Leverage the Wisdom of Crowds to get operator recommendations based on your process design!

Activate Wisdom of Crowds

Parameters

ARIMA

time series attribute

issions (tons)

has indices

indices attribute

sort time series

p: order of the autoregressive model

Hide advanced parameters

Help

ARIMA
Time Series

Tags: Time Series, Forecast, Arima, Arma, Ar, Ma, Forecasting

Synopsis

This operator trains an ARIMA model for a selected time series attribute.

[Jump to Tutorial Process](#)

Description



Views:

Design

Results

Turbo Prep

Auto Model

Interactive
Analysis

Find data, operators...etc

All Studio

Repository

+ Import Data

smoking (5/21/24 1:20 AM - 8 kB)

test (5/5/24 10:19 AM - 143 kB)

train (5/5/24 10:18 AM - 353 kB)

專案1 (5/22/24 5:50 PM - 7 kB)

專案2 (5/22/24 10:27 PM - 3 kB)

專案3 (5/27/24 4:28 AM - 4 kB)

titanic (4/24/24 3:50 AM - 10 kB)

DB (Legacy)

Operators

arima

Modeling (3)

Time Series (3)

Forecasting (2)

Apply Forecast

ARIMA

Utility (1)

Generate Data (ARIMA)

No results were found.

Process

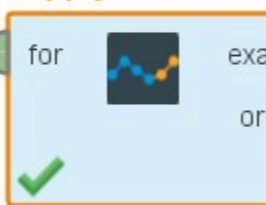
Process

Process

Retrieve 專案_3

ARIMA

Apply Forecast



Parameters

Apply Forecast

forecast horizon 5

☒ add original time series

☒ add combined time series

Help

Apply Forecast
Time Series

Tags: [Time Series](#), [Prediction](#), [Apply Arima](#), [Apply Model](#), [Apply Holt-Winters](#), [Forecasting](#)

Synopsis

This operator uses a forecast model to predict the next values of a time series.

[Jump to Tutorial Process](#)

Leverage the Wisdom of Crowds to get operator recommendations based on your process design!

☒ Activate Wisdom of Crowds

Repository

[+ Import Data](#)

- smoking (5/21/24 1:20 AM - 8 kB)
- test (5/5/24 10:19 AM - 143 kB)
- train (5/5/24 10:18 AM - 353 kB)
- 專案1 (5/22/24 5:50 PM - 7 kB)
- 專案2 (5/22/24 10:27 PM - 3 kB)
- 專案3 (5/27/24 5:28 AM - 6 kB)
- titanic (4/24/24 3:50 AM - 10 kB)
- DB (Legacy)

Operators

FORECAST

- Modeling (7)
 - Time Series (7)
 - Forecasting (6)
 - Apply Forecast
 - ARIMA

We found "Forecasting", "LifeStyle Marketing" and one more result in the Marketplace. [Show me!](#)

Process

Process

Retrieve 專案_3

ARIMA

Apply Forecast

Parameter combination not allowed. The combination of the parameters p,d,q dependent to the length of the series is not allowed. p: 0, d: 0, q: 1, length: 7. For a time series with length 7 the following condition has to be fulfilled for estimating start parameters for the ARIMA fit: $\text{length} > d + \text{Maximum}(p, q+6)$.

Got it!

Parameters

ARIMA

indices attribute: Year

☒ sort time series

p: order of the autoregressive model: 0

d: degree of differencing: 0

q: order of the moving-average model: 1

[Hide advanced parameters](#)

Help

ARIMA

Time Series

Tags: [Time Series](#), [Forecast](#), [Arima](#), [Arma](#), [Ar](#), [Ma](#), [Forecasting](#)

Synopsis

This operator trains an ARIMA model for a selected time series attribute.

[Jump to Tutorial Process](#)

Description

Repository

[+ Import Data](#)

- smoking (5/21/24 1:20 AM - 8 kB)
- test (5/5/24 10:19 AM - 143 kB)
- train (5/5/24 10:18 AM - 353 kB)
- 專案1 (5/22/24 5:50 PM - 7 kB)
- 專案2 (5/22/24 10:27 PM - 3 kB)
- 專案3 (5/27/24 5:28 AM - 6 kB)
- titanic (4/24/24 3:50 AM - 10 kB)
- DB (Legacy)

Operators

performance

- Performance (18)
 - Predictive (7)
 - Performance (Classification)
 - Performance (Binominal Classi
 - Performance (Regression)
 - Performance (Costs)

No results were found.

Process

Process

Retrieve 專案_3

Forecast Validation

Parameter exceeds Time Series...

Sum of window size, horizon size and horizon offset is larger the length of the Time Series. Sum of window size, horizon size and horizon offset is larger the length of the Time Series. Sum of window size, horizon size and horizon offset = 8. Length of Time Series = 7

Got it!

Parameters

Forecast Validation

- ☒ sort time series
- ☐ expert settings
- unit: example based
- window size: 7
- step size: 1
- [Hide advanced parameters](#)
- ☒ [Change compatibility \(10.3.001\)](#)

Help

Forecast Validation

Time Series

Tags: [Time Series](#), [Sliding Window Validation](#), [Evaluations](#), [Performances](#), [Validations](#), [Folds](#), [Validation](#)

Synopsis

This operator performs a validation of a forecast model, which predicts the future values of a time series.

[Jump to Tutorial Process](#)

Country	Row No.	Year	forecast of ...	Carbon Emi...	Carbon Emi...
	1	2016	?	385629	385629
	2	2017	?	379305	379305
	3	2018	?	384130	384130
	4	2019	?	363960	363960
	5	2020	?	304330	304330
	6	2021	?	380330	380330
	7	2022	?	247638	247638
	8	2023	367270.253	?	367270.253
	9	2024	346167.397	?	346167.397
	10	2025	349889.893	?	349889.893
	11	2026	349233.253	?	349233.253
	12	2027	349349.083	?	349349.083

	Row No.	Year	forecast of ...	Fashion & L...	Fashion & L...
	1	2016	?	143336	143336
	2	2017	?	149300	149300
	3	2018	?	153960	153960
	4	2019	?	156830	156830
	5	2020	?	146180	146180
	6	2021	?	145650	145650
	7	2022	?	93677	93677
	8	2023	137179.252	?	137179.252
	9	2024	140923.521	?	140923.521
	10	2025	141245.792	?	141245.792
	11	2026	141273.531	?	141273.531
	12	2027	141275.918	?	141275.918

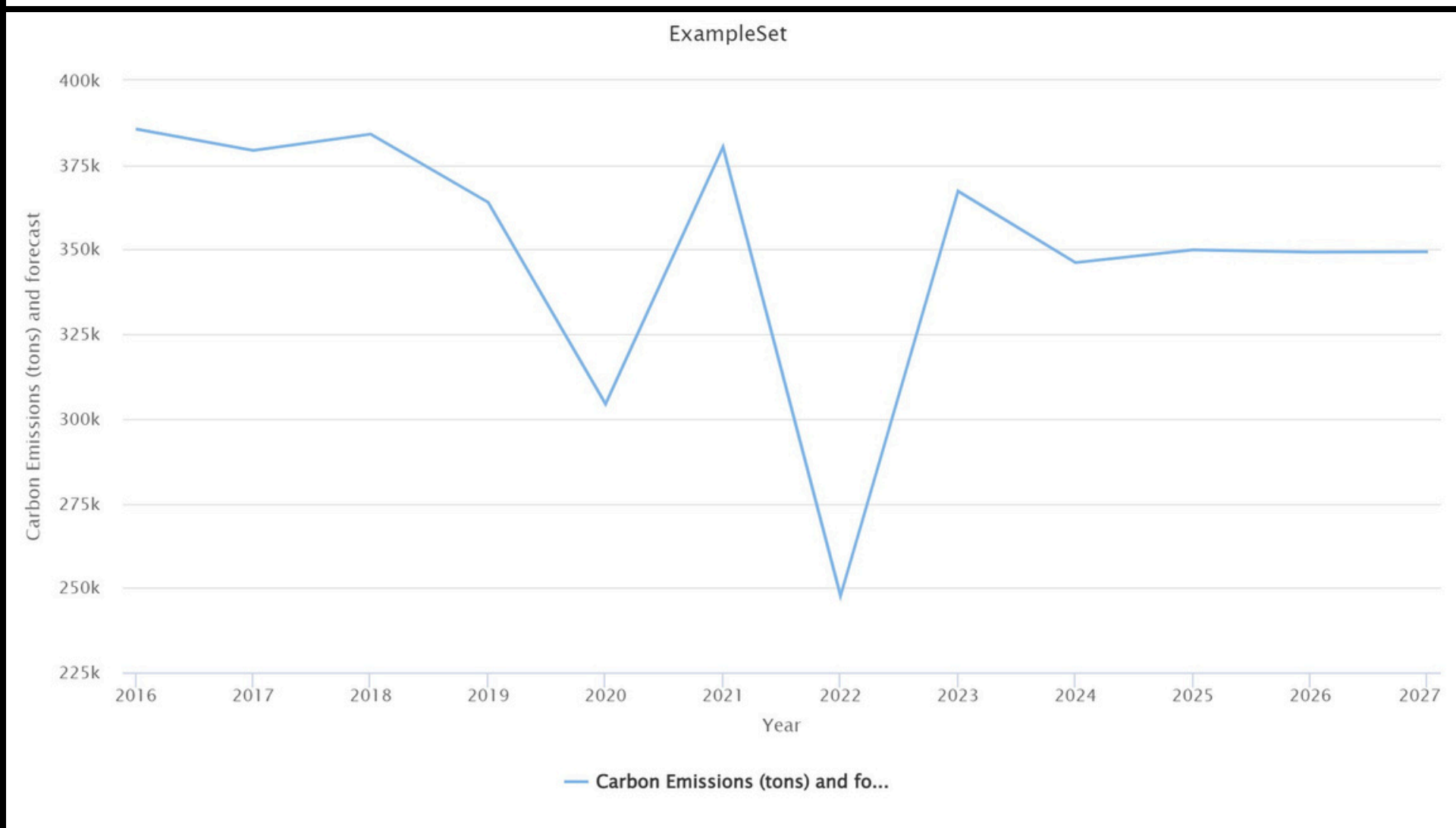
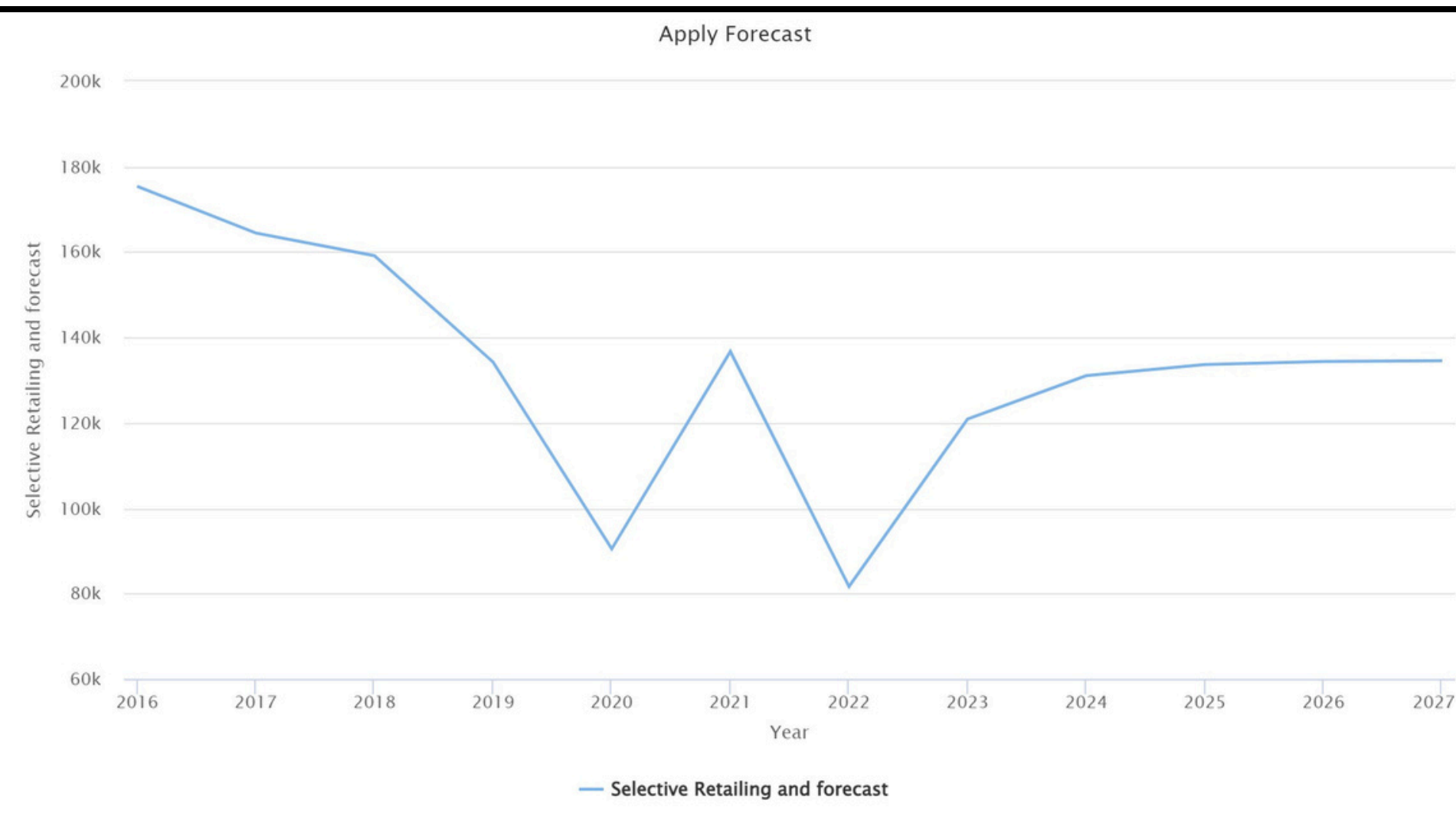
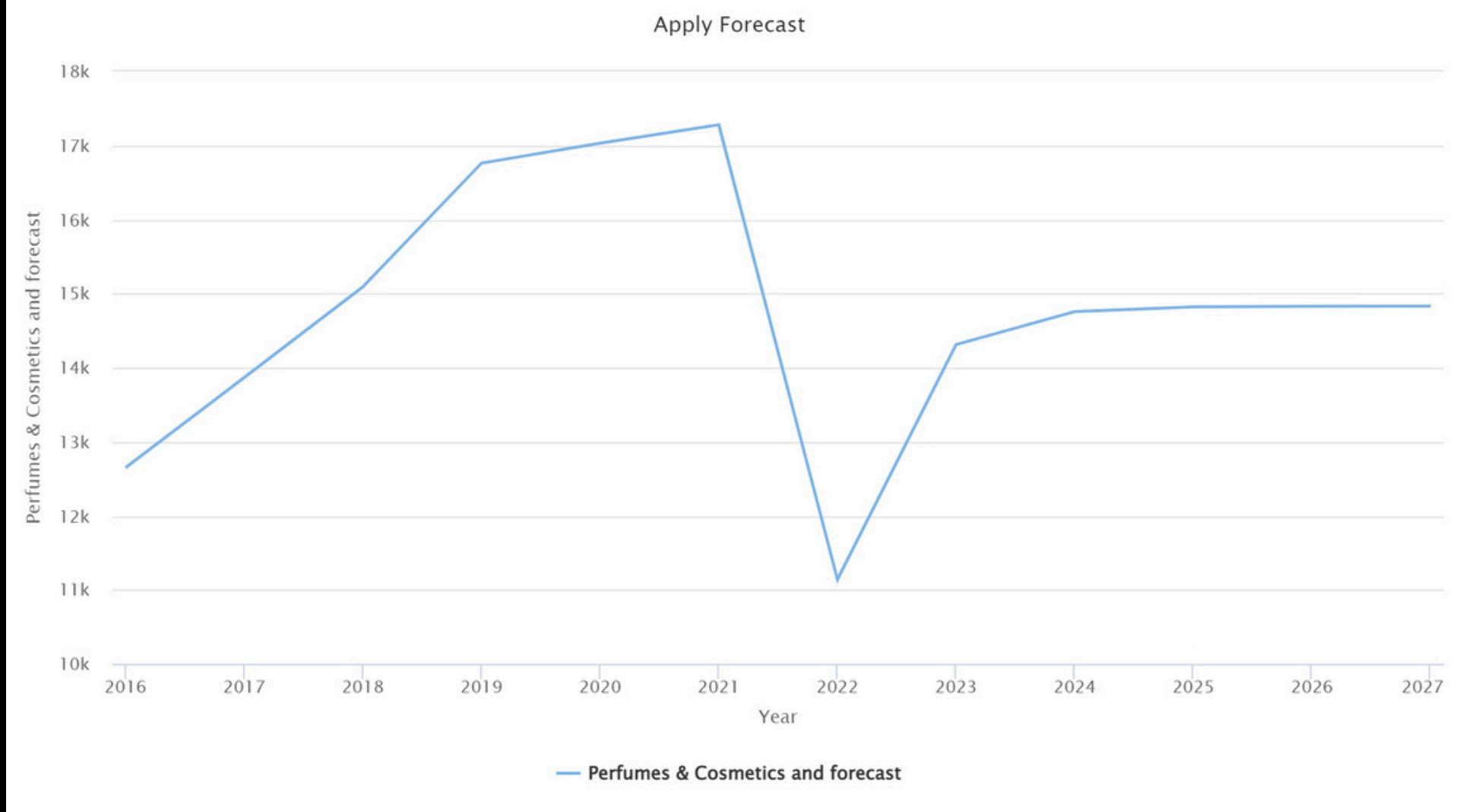
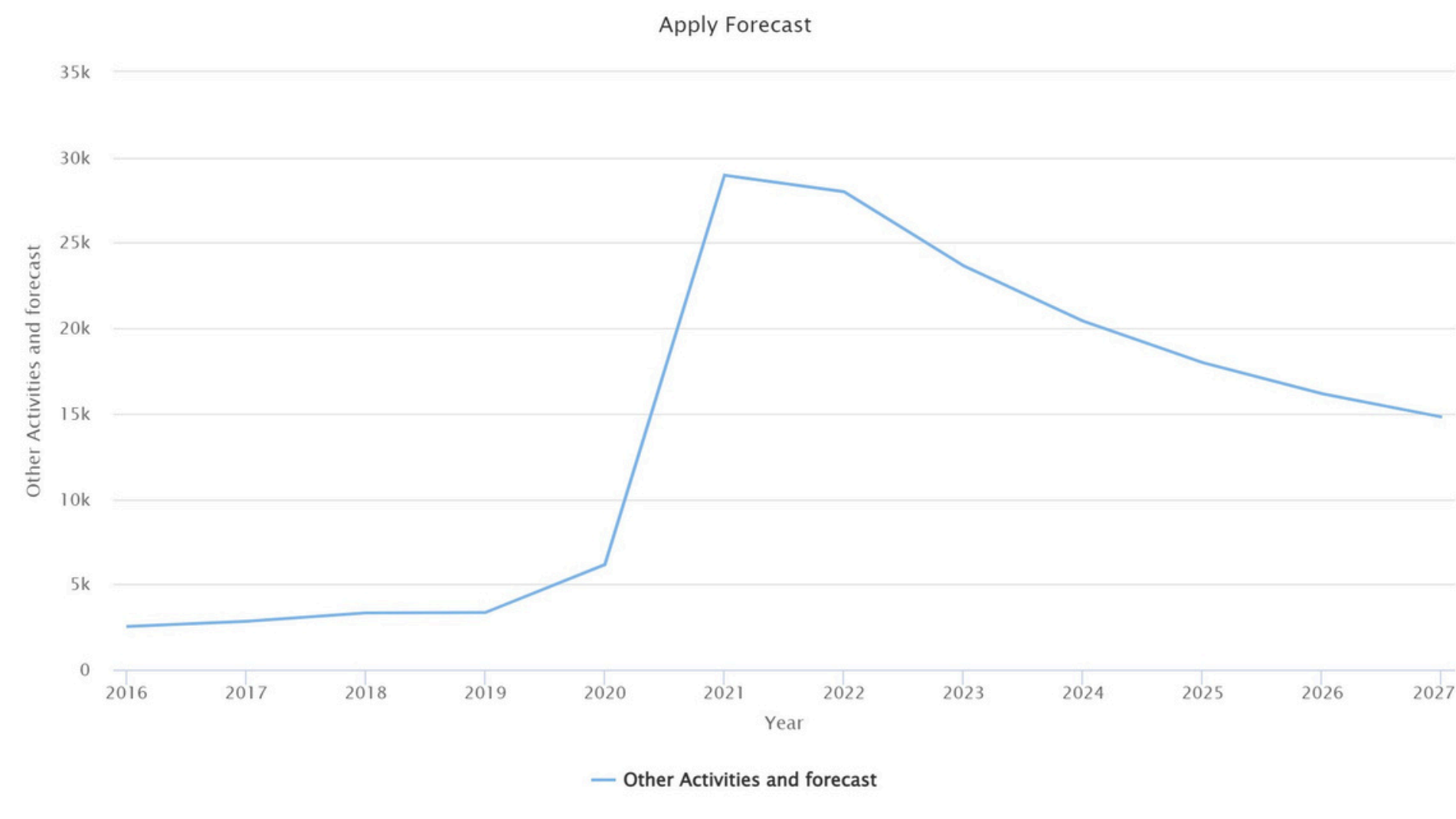
	Row No.	Year	forecast of ...	Other Activi...	Other Activi...
	1	2016	?	2529	2529
	2	2017	?	2830	2830
	3	2018	?	3320	3320
	4	2019	?	3340	3340
	5	2020	?	6150	6150
	6	2021	?	28950	28950
	7	2022	?	27989	27989
	8	2023	23652.649	?	23652.649
	9	2024	20405.795	?	20405.795
	10	2025	17974.705	?	17974.705
	11	2026	16154.420	?	16154.420
	12	2027	14791.477	?	14791.477

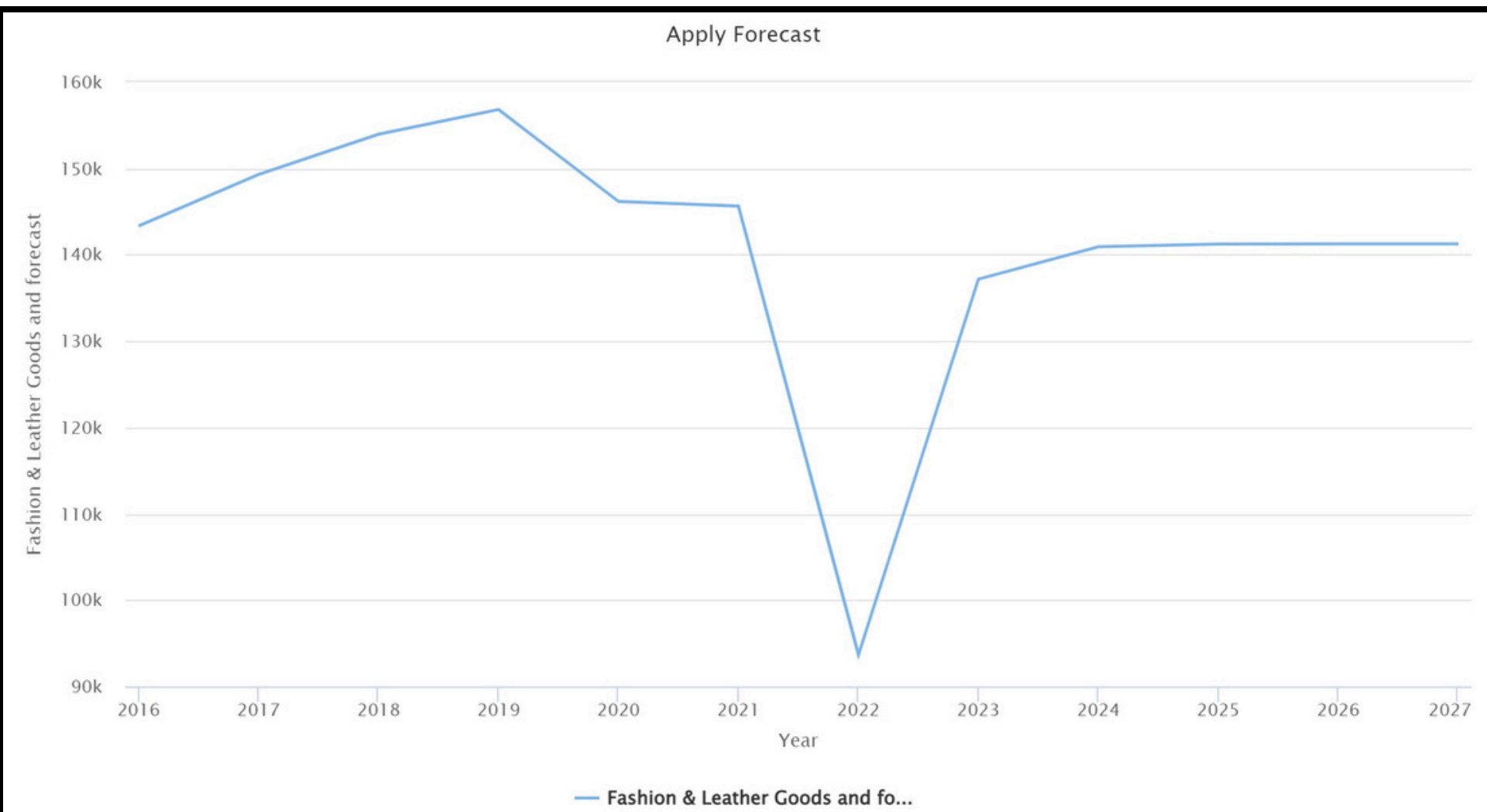
	Row No.	Year	forecast of ...	Perfumes &...	Perfumes &...
	1	2016	?	12650	12650
	2	2017	?	13865	13865
	3	2018	?	15090	15090
	4	2019	?	16760	16760
	5	2020	?	17030	17030
	6	2021	?	17280	17280
	7	2022	?	11140	11140
	8	2023	14310.830	?	14310.830
	9	2024	14757.482	?	14757.482
	10	2025	14820.399	?	14820.399
	11	2026	14829.261	?	14829.261
	12	2027	14830.510	?	14830.510

	Row No.	Year	forecast of ...	Selective R...	Selective R...
	1	2016	?	175377	175377
	2	2017	?	164440	164440
	3	2018	?	159130	159130
	4	2019	?	134230	134230
	5	2020	?	90480	90480
	6	2021	?	136690	136690
	7	2022	?	81650	81650
	8	2023	120883.750	?	120883.750
	9	2024	131030.991	?	131030.991
	10	2025	133655.428	?	133655.428
	11	2026	134334.201	?	134334.201
	12	2027	134509.755	?	134509.755

	Row No.	Year	forecast of ...	Watches & ...	Watches & ...
	1	2016	?	11950	11950
	2	2017	?	12430	12430
	3	2018	?	11780	11780
	4	2019	?	11910	11910
	5	2020	?	10690	10690
	6	2021	?	17290	17290
	7	2022	?	7071	7071
	8	2023	16928.540	?	16928.540
	9	2024	6556.554	?	6556.554
	10	2025	17469.834	?	17469.834
	11	2026	5987.011	?	5987.011
	12	2027	18069.100	?	18069.100

	Row No.	Year	forecast of ...	Wines & Spi...	Wines & Spi...
	1	2016	?	39787	39787
	2	2017	?	36440	36440
	3	2018	?	40850	40850
	4	2019	?	40890	40890
	5	2020	?	33800	33800
	6	2021	?	34470	34470
	7	2022	?	26111	26111
	8	2023	31118.933	?	31118.933
	9	2024	33603.462	?	33603.462
	10	2025	34836.083	?	34836.083
	11	2026	35447.609	?	35447.609
	12	2027	35750.999	?	35750.999







研究結論

- 雖與2026年減少50%的目標仍有些微差距，但整體碳排放量出現下降趨勢。
- 重視供應鏈碳排放的減少，與供應鏈夥伴合作，共同減少碳排放。
- 從預測結果推知，皮革產品的循環服務確實有助減少碳排。因此持續推動產品的修復服務，以及產品的可修復性，可能是集團可以努力的方面。
- 化妝品、香水方面盡可能以自然原料製成，有助豐富生物多樣性。



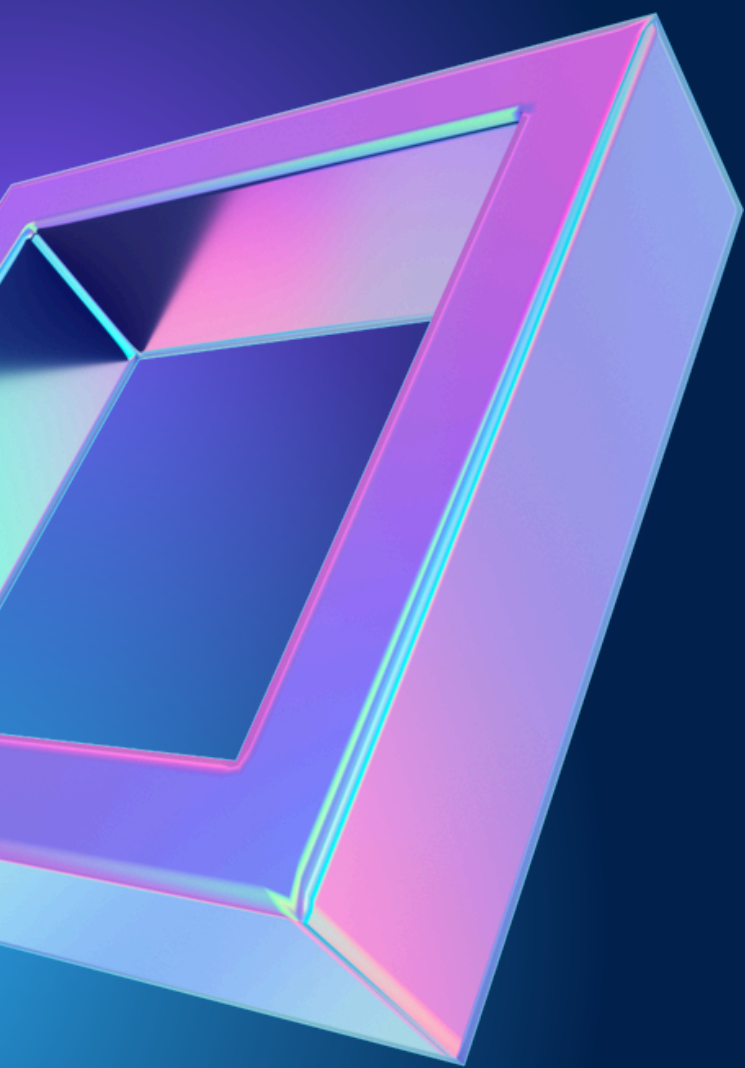
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團隊分工

系級	學號	姓名	工作內容
中文四	109101034	陳靖誼	書面報告整理
教育四	109102023	黃于庭	簡報及海報製作
財政四	109205057	徐祥智	程式設計、簡報製作、 口頭報告
英文三	110501073	嚴維伶	程式設計
經濟二	111208047	王玟晴	程式設計
政治一	112202006	柯知言	書面報告整理



Thank You